

Ricardo (Zhicong) Xie

Auckland, New Zealand | 02902729390 | zxie588@aucklanduni.ac.nz | Portfolio: <https://zxie588-alt.github.io> | GitHub: github.com/zxie588-alt

Product / Hardware Engineering Intern | CAD, Enclosures and Mechatronics

PROFILE

- Mechanical engineering master's student with project evidence across CAD modelling, enclosure design, embedded C, sensor systems and early validation planning.
- Most relevant work: ruggedized IoT sensor module, dynamic weighing firmware, SolidWorks ventilation hardware layout and hands-on manufacturing training.
- Project screenshots and source links: <https://zxie588-alt.github.io>

WORK RIGHTS AND AVAILABILITY

- NZ student visa. Eligible to work up to 25 hours/week during semester and full-time during scheduled university breaks.
- Available full-time from 17 Nov 2026 to late Feb 2027. Auckland preferred; open to NZ relocation for the right internship opportunity.

EDUCATION

University of Auckland - Master of Engineering Studies, Mechanical Engineering | 2026 - Present

Current study/projects in building services, smart manufacturing, CFD/thermal modelling, composite structures and engineering methods.

Huazhong Agricultural University - Bachelor of Engineering, Mechanical Engineering / Mechatronics | 2021 - 2025

Mechanical design, CAD, electrotechnics, sensors, embedded control and mechatronic system integration.

PROJECT WORK

Ruggedized Outdoor IoT Sensor Module

SolidWorks, enclosure design, DFM notes, Keil C, concept validation

- Designed a SolidWorks enclosure concept with screw bosses, gasket stack-up, PCB and battery placeholders, cable/strap mounts and service access.
- Added FEA-style screening for corner load, strap lug and screw boss regions; kept an open test list for sealing, impact, thermal exposure and firmware validation.
- Built concept firmware in Keil C with mock sensor readout, battery monitoring and a simple module state machine; documented the electronics and firmware architecture in the portfolio.

Dynamic Livestock Weighing System

STM32, HX711, embedded C, Bluetooth/serial communication

- Developed STM32 firmware for pressure-sensor acquisition using HX711 hardware and real-time data transmission to an upper computer.
- Supported dynamic weighing logic for improved signal stability under field-like conditions.
- Contributed to a mechatronic system that received Computer Software Copyright Registration No. 2024SR0141888.

NZ Residential MVHR / ERV Ventilation Concept

SolidWorks, AutoCAD/DXF, HVAC sizing, NZBC G4/H1 context, commissioning plan

- Built a residential MVHR/ERV concept around a 3D duct layout, airflow schedule, duct velocity checks and preliminary pressure-loss estimate.
- Modelled the MVHR unit, supply/extract/outdoor/exhaust ducts, diffusers, grilles and hangers in SolidWorks; exported DXF, STEP, STL, PNG and a component list for review.
- Used 60 L/s balanced supply/extract as the normal case and an 85 L/s boost extract check; estimated about 80 Pa normal and 120 Pa boost external static pressure.
- Checked the design against NZBC G4/H1 and Healthy Homes ventilation context; added a commissioning test plan using CO2, humidity, airflow and fan-power measurements.

Manufacturing Workshop Training

Laser cutting, AutoCAD/DXF, CNC machining exposure, 3D printing workflow

- Prepared 2D AutoCAD/DXF geometry for laser cutting and engraving, then compared drawing intent with the physical cut plate.
- Observed and practised CNC machining workflow including workholding, spindle/tool motion, cutting fluid, chip formation and process safety.
- Reviewed machined gear and shaft components to connect CAD features with real manufacturing constraints such as tolerances, edge quality and tool access.

Composite Formula SAE Front Wing Analysis

Abaqus/Standard, shell FEA, composite layup, structural checks

- Built an Abaqus shell model of a composite Formula SAE front wing using S4R elements, laminate definitions and local mesh refinement around attachment regions.
- Converted aero downforce/drag, lateral cornering and 4.1 G bump conditions into structural load cases for strength and stiffness checks.
- Checked ply-level stress utilisation against fatigue and ultimate allowables; the final laminate reached about 74% peak utilisation under the governing ultimate case with SF = 1.5.

TECHNICAL SKILLS

- Product/CAD: SolidWorks assemblies, enclosure features, gasket/screw-boss layout, STEP/STL export, component documentation
- Mechatronics: STM32 C, Keil 5, Proteus, sensors, battery-monitor logic, serial/Bluetooth communication, PCB/circuit basics
- Validation mindset: FEA-style screening, test list writing, DFM notes, firmware state-machine documentation
- Analysis: Abaqus, ANSYS Fluent/CFX, CFD/FEA, heat/mass balance checks
- Documentation: GitHub Pages portfolio, project screenshots, source links, engineering notes

LEADERSHIP AND COMMUNICATION

- Led team-level integration in the orbital data-centre project and commonly take the organiser role in group work.
- Former student news-department lead with photography/event documentation experience; comfortable turning technical work into clear visual evidence.
- Professional working proficiency in English; native Mandarin Chinese.